ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** ask a question	It begins with a question.
2	**(b)** variable	Variables can change.
3	**(d)** number and unit	e.g. 5 m.
4	**(d)** thermometer	Thermometer measures temperature.
5	**(a)** one	Only one.
6	**(c)** quantitative observation	It contains a number.
7	**(b)** stopwatch	A stopwatch measures time.
8	**(d)** wear goggles while heating	Goggles protect eyes.
9	**(d)** qualitative observation	No number is given.
10	**(b)** balance	A balance measures mass.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A quantity or factor that can change (2).	2
12	A testable guess or explanation (2).	2
13	To check reliability (1); To reduce chance errors (1).	1+1
14	A test in which only one factor is changed (1); All other factors are kept the same (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Question and hypothesis (1); Test, observe and record (1); Analyse and conclude (1).	1+1+1
16	Length of a pencil 15 cm (1); Mass of a book 500 g (1); Temperature 30 degrees Celsius (1).	1+1+1
17	Only one factor is changed (1); e.g. same size of puri while changing oil temperature (1); So that the result can be explained (1).	1+1+1
	OR: Metre scale: length (1); Balance: mass (1); Thermometer: temperature (1).	1+1+1
18	Colour of a flower (1); Smell of a substance (1); Taste of a fruit (1).	1+1+1
	OR: Date and title of experiment (1); Observations in a table (1); Conclusion (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Question: why does bread rise? (1); Hypothesis: yeast makes gas (1); Experiment: bread with and without yeast (1); Observe and record (1); Conclusion (1).	1+1+1+1+1
	OR: Qualitative: no number (1); Quantitative: with number (1); Qualitative examples: colour, smell, taste (1); Quantitative: mass, time, length (1); Both are useful in science (1).	1+1+1+1+1
20	Independent changes by us (1); Dependent is measured (1); Constant stays same (1); Example: plant growth with water (1); Water amount is independent; height is dependent (1).	1+1+1+1+1
	OR: Same type of plants (1); Same amount of water and soil (1); One pot in light, one in dark (1); Measure height each day (1); Compare and conclude (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up the **independent** variable (changed by you) with the **dependent** variable (measured).
- Changing more than one factor at a time and still calling the test fair.
- Writing a hypothesis as a question or an opinion instead of a testable statement.
- Taking a single reading and skipping repeats; forgetting to average readings.
- Reading a scale or thermometer from the side (parallax) and omitting units in the answer.
- Drawing a conclusion that the data do not support, or generalising from one trial.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** away from people	Hot liquid may splash out.
2	**(a)** check and use them	Results are verified.
3	**(a)** metre scale	Scale gives length.
4	**(b)** kilogram	Mass is in kg.
5	**(d)** show data visually	Graphs picture data.
6	**(d)** observation	Observations are noted down.
7	**(b)** metre	Length is in m.
8	**(a)** 6	(4+6+8)/3 = 6.
9	**(d)** hypothesis	Hypothesis is testable.
10	**(a)** control	A control stays unchanged.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Wear goggles and apron (1); Never taste or smell chemicals directly (1).	1+1
12	Qualitative: it is colourless (1); Quantitative: boils at 100 degrees Celsius (1).	1+1
13	(a) Thermometer (1); (b) Stopwatch (1).	1+1
14	To keep it organised (1); To compare easily (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum = 42 s (1); Divide by 3 (1); Average = 14 s (1).	1+1+1
16	It would be dishonest (1); It gives a false conclusion (1); Others who rely on it may be harmed (1).	1+1+1
17	Measuring cup (volume) (1); Kitchen weighing scale (mass) (1); Timer (time) (1).	1+1+1
	OR: Keep flammable things away (1); Tie hair back (1); Switch off or close the gas after use (1).	1+1+1
18	Keep the bulb in the water, not touching the vessel (1); Wait until the reading is steady (1); Read with the eye level with the mark (1).	1+1+1
	OR: Guess is not exact (1); Different people guess differently (1); Stopwatch gives accurate value (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Real problems are complex (1); Scientists may repeat steps (1); Results may lead to new questions (1); Order can change (1); Flexibility helps discovery (1).	1+1+1+1+1
	OR: Others rely on the data (1); False data misleads (1); Errors cannot be corrected (1); Trust in science falls (1); Honesty builds knowledge (1).	1+1+1+1+1
20	Wear goggles (1); Tie hair (1); Do not taste chemicals (1); Handle hot things with tongs (1); Inform the teacher about accidents (1).	1+1+1+1+1
	OR: Metre rule: length (1); Balance: mass (1); Stopwatch: time (1); Thermometer: temperature (1); Measuring cylinder: volume (1).	1+1+1+1+1

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Exploring the Investigative World of Science** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** curious and honest	Honesty and curiosity matter.
2	**(c)** record data neatly	Tables organise data.
3	**(d)** Does salt dissolve faster in warm water?	It can be tested.
4	**(d)** The liquid is sweet.	Taste is a quality.
5	**(a)** conclusion	Conclusion gives the answer.
6	**(a)** reliable results	More trials give more reliable results.
7	**(b)** measuring cylinder	Cylinder measures volume.
8	**(b)** second	SI unit of time is second.
9	**(c)** parallax	Eye must be in line with the mark.
10	**(a)** observing and questioning	Observation starts it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Statement of the answer from the data (2).	2
12	Information gathered with senses or instruments (2).	2
13	(a) metre (1); (b) kilogram (1).	1+1
14	Curiosity (1); Honesty (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Independent: room temperature (1); Dependent: time for ice to melt (1); Constant: size/mass of ice (1).	1+1+1
16	So that measurements are same all over the world (1); metre, kilogram, second (2).	1+2
17	Careful observation (1); Asking questions (1); Recording and sharing results (1).	1+1+1
	OR: Observation is what is seen or measured (1); Conclusion is the answer drawn from observations (1); Conclusion comes after the observations (1).	1+1+1
18	Gives tools and machines (1); Gives medicines (1); Helps us understand nature (1).	1+1+1
	OR: Side viewing causes parallax (1); Reading becomes more or less than the true value (1); Straight view gives correct value (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Apparent shift of reading when viewed from an angle (1); Seen with a scale or a thermometer (1); Keep the eye in line with the mark (1); Look straight on (1); Gives correct reading (1).	1+1+1+1+1
	OR: Repeat reduces chance errors (1); Averaging gives a typical value (1); Sum = 18 (1); Average = 6 s (1); More trials, more confidence (1).	1+1+1+1+1
20	Tables: organise (1); Tables: compare (1); Graphs: show trends (1); Graphs: easier to predict (1); Both aid analysis (1).	1+1+1+1+1
	OR: Choose a testable question (1); Form a hypothesis (1); Identify variables (1); Collect data (1); Draw a conclusion (1).	1+1+1+1+1

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